Extreme Estimation

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Outline

1. Introduction and examples.
2. Consistency
3. Asymptotic normality
4. Estimating the asymptotic variance
5. Inference
6. Two-step M-estimators

Introduction and examples

Motivation for extreme estimation framework: lack of "closed form"

- Both the OLS and IV estimators have "closed form", e.g. $\hat{\beta}_{ols} = (X'X)^{-1}(X'Y)$. That is, given data, the estimates can be computed using a known formula.
- Closed form makes the asymptotic properties of an estimator easy to derive.
- For nonlinear models, estimators usually do not have closed form.
- Even if the underlying model is linear, estimators may do not have closed form.

Median regression example

- Consider median regression,

$$\mathrm{Med}(y|x) = x'\theta = \theta_1 + \theta_2 x_2 + \cdots + \theta_K x_K.$$

  Here $\mathrm{Med}(y|x)$ is the conditional *median* of y given x.

- Without additional assumptions, OLS does not consistently estimate the θ_j. Consider a simple case $\mathrm{Med}(y|x) = \theta_1$. The OLS converges to $\mathrm{E}(y)$, which does not have to be its median.

- But we will show *least absolute deviations* (LAD) does. Let $\hat{Q}_n(b) = n^{-1}\sum_{i=1}^{n}|y_i - x_i'b|$. The LAD estimator solves the minimization problem,

$$\min_{b\in\mathbb{R}^{K+1}} \hat{Q}_n(b). \tag{1}$$

  This problem does not have closed form solution.

- The difficulty in deriving the asymptotic properties of the LAD estimator is the lack of closed form.

- Let $\hat{\theta}_{LAD}$ be a solution of eq. (1). In principle, we can write

$$\hat{\theta}_{LAD} = g_n(z_1, \ldots, z_n), \quad z_i = (x_i', y_i)',$$

  for some function g_n. The problem is we do not know g_n.
- Question: If we do not know g_n, how do we study the large-sample (asymptotic) properties of $\hat{\theta}_{LAD}$?
- Extreme estimation theory provides a framework of studying the asymptotic property of an estimator based on the objective function.

- It starts with the observation that the sample objective function $\hat{Q}_n(b)$ usually converges to some population objective function $Q_0(b)$ that is minimized at the true value.
- For each b, $\hat{Q}_n(b) \equiv n^{-1} \sum_{i=1}^{n} |y_i - x_i' b|$ is an average of iid random function, $q(z_i, b) \equiv |y_i - x_i' b|$, for which we can apply the LLN.
- It would not be a surprise that $\hat{Q}_n(b) \to_p Q_0(b) \equiv \mathrm{E}(q(z_i, b))$ for all b.
- The true θ_0 also minimizes $Q_0(b)$.
- If $\hat{Q}_n(\theta) \to_p Q_0(\theta)$ for all θ, it is not crazy to expect that $\hat{\theta}_n \to_p \theta_0$.

Nonlinear least square example

- Suppose for $y \geq 0$, and we specify an exponential conditional mean model:

$$\mathrm{E}(y \mid x) = \exp(x'\theta).$$

- OLS using $\log(y)$ as the dependent variable does not work, if $\Pr(y = 0) > 0$.
- Letting $\hat{Q}_n(b) = n^{-1} \sum_{i=1}^{n} (y_i - \exp(x_i' b))^2$, *nonlinear least squares* (NLS) estimator $\hat{\theta}_{nls}$ solves

$$\min_b \hat{Q}_n(b).$$

- NLS $\hat{\theta}_{nls}$ does not have a closed form. But it minimizes a function that is an average of i.i.d. random functions of b too.
- We expect $\hat{Q}_n(\theta)$ converges to $Q_0(\theta) = \mathrm{E}([y_i - \exp(x_i'\theta)]^2)$.

- We have the following useful result: assume $\mathrm{E}(y^2) < \infty$ and let $\mathrm{E}(y \mid x) = \mu_0(x)$. Then among all functions $\mu(x)$ with $\mathrm{E}\big(\mu(x)^2\big) < \infty$,

$$\mathrm{E}([y - \mu_0(x)]^2) \leq \mathrm{E}([y - \mu(x)]^2)$$

  It follows that if $\mathrm{E}(y \mid x) = m(x, \theta_0)$, then

$$\mathrm{E}([y - m(x, \theta_0)]^2) \leq \mathrm{E}([y - m(x, \theta)]^2), \text{ all } \theta \in \Theta.$$

- That is, $Q_0(\theta) = \mathrm{E}([y - m(x, \theta_0)]^2)$ is minimized at θ_0.
- So we hope $\hat{\theta}_{nls}$ converges to θ_0.

GMM example

- Suppose that there is a "moment function" vector $g(z, \theta)$ such that $\mathrm{E}(g(z, \theta_0)) = 0$ for some constants θ_0.
- A GMM estimator aims to estimate θ_0 by minimizing

$$\min_{\theta} \hat{Q}_n(\theta) \equiv= \left(n^{-1} \sum_{i=1}^{n} g(z_i, \theta) \right)' \hat{W} \left(n^{-1} \sum_{i=1}^{n} g(z_i, \theta) \right),$$

  where $\hat{W}$ is a weighting matrix.
- Excepting for special g, we do not have a closed form solution of the above problem.
- It would not be a surprise that

$$\hat{Q}_n(\theta) \to_p Q_0(\theta) \equiv (\mathrm{E}(g(z_i, \theta)))' W_0 (\mathrm{E}(g(z_i, \theta))).$$

Consistency of extreme estimators

Notation

- Let θ denote a finitely dimensional generic vector of parameters, and let θ_0 denote the true value.
- Extreme estimator $\hat{\theta}_n$ of θ_0 is from

$$\hat{\theta}_n = \arg\min_{\theta \in \Theta} \hat{Q}_n(\theta).$$

  Θ is the parameter space, which contains all possible candidates of θ_0.
- If $\hat{Q}_n(\theta)$ has a sample average form, that is if $\hat{Q}_n(\theta) = n^{-1} \sum_{i=1}^{n} q(z_i, \theta)$ for some q, the estimator is typically called M estimator.
- In the next few lectures, we use M estimation theory frequently.

Consistency theorem

Theorem 1 *Let* $\left\{\hat{Q}_n(\theta)\right\}$ *be a sequence of random functions on a compact set* $\theta \subset \mathbb{R}^p$ *such that for a continuous real function* $Q_0(\theta)$ *on* θ,

$$\hat{Q}_n(\theta) \to Q_0(\theta) \quad \textit{a.s. (in pr.)} \textbf{ uniformly } \textit{on } \Theta.$$

(convergence of criterion function)

Let $\hat{\theta}_n$ *be any random vector in* Θ *satisfying*

$$\hat{Q}_n(\hat{\theta}_n) = \inf_{\theta \in \Theta} \hat{Q}_n(\theta) \qquad \text{(extreme in sample criterion)}$$

and let θ_0 *be a* **unique** *point in* Θ *such that*

$$Q_0(\theta_0) = \inf_{\theta \in \Theta} Q_0(\theta). \qquad \text{(extreme in population criterion)}$$

Then $\hat{\theta}_n \to \theta_0$ *a.s. (in pr.)*

Uniform convergence in probability: $\hat{Q}_n(\theta)$ converges uniformly in probability to $Q_0(\theta)$ means

$$\sup_{\theta \in \Theta} \left| \hat{Q}_n(\theta) - Q_0(\theta) \right| \to_p 0.$$

Sketch of proof

- First, we show $Q_0(\hat{\theta}_n) - Q_0(\theta_0) \to_p 0$.

$$
\begin{aligned}
0 &\leq Q_0(\hat{\theta}_n) - Q_0(\theta_0) \\
&= Q_0(\hat{\theta}_n) - \hat{Q}_n(\hat{\theta}_n) + \hat{Q}_n(\hat{\theta}_n) - Q_0(\theta_0) \\
&\leq Q_0(\hat{\theta}_n) - \hat{Q}_n(\hat{\theta}_n) + \hat{Q}_n(\theta_0) - Q_0(\theta_0) \\
&\leq 2 \cdot \sup_{\theta \in \Theta} \left| \hat{Q}_n(\theta) - Q_0(\theta) \right| = o_p(1).
\end{aligned}
$$

- By continuity of $Q_0(\cdot)$, $Q_0(\hat{\theta}_n) - Q_0(\theta_0) \to_p 0$ implies $\hat{\theta}_n \to_p \theta_0$.

Uniform convergence

- When $\hat{Q}_n(\theta)$ is sample average, the *uniform law of large numbers* (ULLN) is a powerful device for establishing the uniform convergence.
- Here is the ULLN from Lemma 2.4 of Newey-McFadden.

  Lemma 1 *If the data are iid, Θ is compact, $a(z_i, \theta)$ is continuous at each $\theta \in \Theta$ with probability one, and there is $d(z)$ with $\|a(z, \theta)\| \leq d(z)$ for all $\theta \in \Theta$ and $\mathrm{E}(d(z)) < \infty$, then $\mathrm{E}(a(z, \theta))$ is continuous and $\sup_{\theta \in \Theta} \|n^{-1} \sum_{i=1}^{n} a(z_i, \theta) - \mathrm{E}(a(z, \theta))\| \to_p 0$.*
- ULLN is enough for many examples of extreme estimator, including MLE, NLS, and GMM.
- A more general theorem is in Newey (1991, Uniform Convergence in Probability and Stochastic Equicontinuity, *Econometrica*), which shows that compactness, pointwise convergence and stochastic equicontinuity implies uniform convergence.

"Uniquely minimized at θ_0" is an identification condition

Example 1 *Suppose $x > 0$ is a scalar, $y \geq 0$, and $m(x, \theta) = \theta_1 + \theta_2 x^{\theta_3}$, where the parameter spaces is $\Theta = [0, \infty) \times [0, \infty) \times \mathbb{R}$. If $\theta_{02} = 0$, $\mathrm{E}(y \mid x) = \mathrm{E}(y) = \theta_{01}$, then any θ of the form $(\theta_{01}, 0, \theta_3)'$ minimizes the mean squared error. What would NLS estimate for θ_{03}?*

- We cannot identify θ_{03}.

Identification assumption for NLS

- We need an identification assumption for NLS that rules out the previous situation.

  Assumption 1 *Let* $m(x, \theta) = \mathrm{E}(y \mid x)$*, and* $\mathrm{E}([m(x, \theta_0) - m(x, \theta)]^2) > 0$ *for all* $\theta \in \Theta$*,* $\theta \neq \theta_0$*.*

- Assumption 1 plays the role of a rank condition. In the linear case, $m(x, \theta) = x'\theta$, and then

$$m(x, \theta_0) - m(x, \theta) = (\theta_0 - \theta)' x x' (\theta_0 - \theta).$$

$$\mathrm{E}\Big((m(x, \theta_0) - m(x, \theta))^2\Big) = (\theta_0 - \theta)' \, \mathrm{E}(xx')(\theta_0 - \theta).$$

  For the last expression to be positive for all $\theta \neq \theta_0$, we need $\mathrm{E}(xx')$ to have full rank.

Convergence of estimated sample averages

- Lemma 4.3 of Newey and McFadden is a very useful result for estimating asymptotic variance.

  Lemma 2 *If z_i is iid, $a(z, \theta)$ is continuous at θ_0 with probability one, and there is a neighborhood $\mathcal{N}$ of θ_0 such that $\mathrm{E}(\sup_{\theta \in \mathcal{N}} \|a(z, \theta)\|) < \infty$, then for any $\hat{\theta}_n \to_p \theta_0$,*

$$n^{-1} \sum_{i=1}^{n} a(z_i, \hat{\theta}_n) \to_p \mathrm{E}(a(z, \theta_0)).$$

- This lemma can be used to show the heteroskedasticity robust variance estimator in linear regression.

Asymptotic normality of extreme estimator

Main result

Theorem 2 *Let the conditions of theorem 1 hold and assume in addition that:*

1. Θ *is convex and* θ_0 *is an interior point of* Θ*;*
2. *the first order and second order derivatives of* $\hat{Q}_n(\theta)$ *are well defined as vector and matrix of random functions on* Θ*;*
3. $\sqrt{n}\partial\hat{Q}_n(\theta_0)/\partial\theta \to_d \mathcal{N}(0, B)$*, where* B *is a positive semi-definite matrix;*
4. *for* i, j*,* $\partial^2\hat{Q}_n(\theta)/\partial\theta_i\partial\theta_j \to_p \partial^2 Q_0(\theta)/\partial\theta_i\partial\theta_j$ *uniformly on a neighborhood of* θ_0*, where the limit function involved is continuous in* θ_0*;*
5. *the matrix* $A = \partial^2 Q_0(\theta_0)/\partial\theta\partial\theta'$ *is non-singular.*

Then $\sqrt{n}(\hat{\theta}_n - \theta_0) \to_d \mathcal{N}(A^{-1}BA^{-1})$.

Useful matrix concept

- Let θ be p dimensional vector. Let $a(\theta)$ be a real valued function of θ. Let $f^k(\theta) = (f_1(\theta), \ldots, f_k(\theta))'$ be a k dimensional vector of functions of θ.
- *Gradient* is a $p \times 1$ vector:

$$\nabla a = (\partial a/\partial\theta_1, \partial a/\partial\theta_2, \ldots, \partial a/\partial\theta_k)'.$$

- *Jacobian* matrix is a $k \times p$ matrix, that is $\dim(f) \times \dim(\theta)$:

$$\nabla f = \begin{bmatrix} \partial f_1/\partial\theta_1 & \dots & \partial f_1/\partial\theta_p \\ \vdots & \vdots & \vdots \\ \partial f_k/\partial\theta_1 & \dots & \partial f_k/\partial\theta_p \end{bmatrix}$$

- *Hessian* matrix is a $p \times p$ matrix, that is $\dim(\theta) \times \dim(\theta)$:

$$\begin{bmatrix} \partial^2 a/\partial^2\theta_1 & \dots & \partial^2 a/\partial\theta_1\partial\theta_p \\ \vdots & \vdots & \vdots \\ \partial^2 a/\partial\theta_p\partial\theta_1 & \dots & \partial^2 a/\partial^2\theta_p \end{bmatrix}$$

- Hessian matrix is the Jacobian of gradient.

Proof of asymptotic normality in M estimation case

- We show the proof in M estimation case.
- In the M estimation, we minimize an objective function $\hat{Q}_n(\theta) = n^{-1} \sum_{i=1}^{n} q(z_i, \theta)$.
- For the minimization problem, we are interested in its FOC. Let $s(z, \theta) = \nabla_\theta q(z, \theta)$ be the gradient of q w.r.t. θ.
- $s(z, \theta)$ is often called **score function** following the convention of MLE.
- Because θ_0 is in the interior of Θ and $\hat{\theta}_n \to_p \theta_0$, we know that $\hat{\theta}_n$ falls in the interior of Θ with probability one.
- Since $\hat{\theta}_n \in \mathrm{int}(\Theta)$, as a minimizer, $\hat{\theta}_n$ satisfies

$$\sum_{i=1}^{n} s(z_i, \hat{\theta}_n) = 0.$$

  This is a set of $p = \dim(\theta)$ equations.

- Assuming that q is twice continuously differentiable, each component of $s(z, \theta)$ is continuously differentiable.
- We then can apply the mean value theorem,

$$0 = \sum_{i=1}^{n} s(z_i, \hat{\theta}_n) = \sum_{i=1}^{n} s(z_i, \theta_0) + \left(\sum_{i=1}^{n} \nabla_\theta s(z_i, \theta_*) \right) (\hat{\theta}_n - \theta_0).$$

  Here θ_* is on the line segment between $\hat{\theta}_n$ and θ_0, hence $\theta_* \to_p \theta_0$.
- Write the Hessian matrix

$$H(z, \theta) = \nabla_\theta s(z, \theta) = \frac{\partial^2 q(z, \theta)}{\partial \theta \partial \theta'}$$

- Using the Hessian matrix notation, we have

$$0 = \sum_{i=1}^{n} s(z_i, \hat{\theta}_n) = \sum_{i=1}^{n} s(z_i, \theta_0) + \left(\sum_{i=1}^{n} H(z_i, \theta_*) \right) (\hat{\theta}_n - \theta_0)$$

- So

$$0 = n^{-1/2} \sum_{i=1}^{n} s(z_i, \theta_0) + \left(n^{-1} \sum_{i=1}^{n} H(z_i, \theta_*) \right) \sqrt{n}(\hat{\theta}_n - \theta_0)$$

- By $\theta_* \to_p \theta_0$, we conclude

$$n^{-1} \sum_{i=1}^{n} H(z_i, \theta_*) \to_p \mathrm{E}(H(z, \theta_0)) \equiv A,$$

by lemma 2.

- Assuming A is nonsingular (hence $n^{-1}\sum_{i=1}^{n} H(z_i, \theta_*)$ is nonsingular w.p.a.1), we write

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \left(n^{-1}\sum_{i=1}^{n} H(z_i, \theta_*)\right)^{-1}\left(-n^{-1/2}\sum_{i=1}^{n} s(z_i, \theta_0)\right)$$

- By the CLT,

$$-n^{-1/2}\sum_{i=1}^{n} s(z_i, \theta_0) \to_d \mathcal{N}(-\mathrm{E}(s(z, \theta_0)), \mathrm{Var}(s(z, \theta_0))).$$

- By the interchangeability of derivative and expectation (integral) and the FOC, we have

$$\mathrm{E}(s(z, \theta_0)) = \nabla_\theta \mathrm{E}(q(z, \theta_0)) = 0.$$

- Similarly, we have A is positive definite,

$$A = \mathrm{E}\left(\frac{\partial^2 q(z, \theta_0)}{\partial\theta\partial\theta'}\right) = \frac{\partial^2 \mathrm{E}(q(z, \theta_0))}{\partial\theta\partial\theta'}.$$

- Letting $B = \operatorname{Var} s(z, \theta_0) = \mathrm{E}(s(z, \theta_0) s(z, \theta_0)')$, we conclude

$$n^{-1/2} \sum_{i=1}^{n} s(z_i, \theta_0) \to_d \mathcal{N}(0, B).$$

- By the Slusky lemma, we conclude

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \to_d A^{-1} \left(n^{-1/2} \sum_{i=1}^{n} s(z_i, \theta_0) \right)$$

  hence

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \to_d \mathcal{N}(0, A^{-1} B A^{-1}).$$

- The asymptotic variance of $\hat{\theta}_n$ is $n^{-1} A^{-1} B A^{-1}$, which depends on the expected value of Hessian matrix of $q(z, \theta)$, the variance of score $\nabla_\theta q(z, \theta)$ (both evaluated at θ_0), and the sample size.

Estimating the asymptotic variance

Components of asymptotic variance

- The asymptotic variance of $\hat{\theta}_n$ is $n^{-1}A^{-1}BA^{-1}$, where

$$A = \mathrm{E}(H(z, \theta_0)) = \mathrm{E}\left(\frac{\partial^2 q(z, \theta_0)}{\partial\theta\partial\theta'}\right),$$

$$B = \mathrm{E}(s(z, \theta_0)s(z, \theta_0)') = \mathrm{E}\left(\frac{\partial q(z, \theta_0)}{\partial\theta}\frac{\partial q(z, \theta_0)}{\partial\theta}'\right).$$

- By lemma 2, we can estimate A and B by

$$\hat{A} = n^{-1}\sum_{i=1}^{n} H(z_i, \hat{\theta}_n),$$

$$\hat{B} = n^{-1}\sum_{i=1}^{n} s(z_i, \hat{\theta}_n)s(z_i, \hat{\theta}_n)'.$$

- Usually, $z_i = (y_i, x_i')'$, e.g. NLS. We can write

$$A = \mathrm{E}(H(z, \theta_0)) = \mathrm{E}(\mathrm{E}(H(z, \theta_0) \mid x)).$$

Let $A(x, \theta_0) \equiv \mathrm{E}(H(z, \theta_0) \mid x)$. In some cases, $A(x, \theta_0)$ might be easier to estimate. Let $A(x, \hat{\theta}_n)$ be the estimator. Then another estimator of A is

$$\hat{A} = n^{-1} \sum_{i=1}^{n} A(x_i, \hat{\theta}_n).$$

- The asymptotic variance estimator is then

$$n^{-1} \hat{A}^{-1} \hat{B} \hat{A}^{-1} = \left(\sum_{i=1}^{n} H(z_i, \hat{\theta}_n) \right)^{-1} \left(\sum_{i=1}^{n} s(z_i, \hat{\theta}_n) s(z_i, \hat{\theta}_n)' \right) \left(\sum_{i=1}^{n} H(z_i, \hat{\theta}_n) \right)^{-1}$$

Asymptotic variance: NLS example

- For NLS $\mathrm{E}(y \mid x) = m(x, \theta)$, we let $z = (y, x)$ and $q(z, \theta) = (y - m(x, \theta))^2/2$.
- Score:

$$s(z, \theta) = -\nabla_\theta m(x, \theta)(y - m(x, \theta))$$

  Clearly, $\mathrm{E}(s(z, \theta_0)) = 0$.
- Hessian:

$$H(z, \theta) = \nabla_\theta s(z, \theta) = \nabla_\theta m(x, \theta) \nabla_\theta m(x, \theta)' - \nabla_\theta^2 m(x, \theta) u(\theta),$$

  where $u(\theta) = y - m(x, \theta)$.

- Note that $\mathrm{E}(u(\theta_0) \mid x) = 0$. We can estimate $A = \mathrm{E}(H(z, \theta_0))$ by

$$\hat{A} = n^{-1} \sum_{i=1}^{n} \nabla_\theta m(x_i, \hat{\theta}_n) \nabla_\theta m(x_i, \hat{\theta}_n)'$$

- For B, we have

$$\hat{B} = n^{-1} \sum_{i=1}^{n} \hat{u}_i^2 \nabla_\theta m(x_i, \hat{\theta}_n) m(x_i, \hat{\theta}_n)'.$$

- We then have the following heteroskedasticity robust asymptotic variance estimator for the NLS,

$$\left(\sum_{i=1}^{n} \nabla_\theta \hat{m}_i \nabla_\theta \hat{m}_i' \right)^{-1} \left(\sum_{i=1}^{n} \hat{u}_i^2 \nabla_\theta \hat{m}_i \nabla_\theta \hat{m}_i' \right) \left(\sum_{i=1}^{n} \nabla_\theta \hat{m}_i \nabla_\theta \hat{m}_i' \right)^{-1}$$

  Here $\hat{m}_i \equiv m(x_i, \hat{\theta}_n)$.

The bootstrap

- As you have seen the asymptotic variance sometimes can be difficult to compute. The bootstrap is a convenient method for estimating the variance and the distribution of a statistic when the variance formula is difficult to compute.
- Suppose we have iid sample $x_1, \ldots, x_n$ and a test statistic $T_n = g(x_1, \ldots, x_n)$, e.g. an estimator. Let F denote the CDF x.
- Let $v_F(T_n)$ denote the variance of T_n. We have added the subscript F to emphasize that the variance is a function of F. If we knew F, we could at least in principle compute the variance. For example, $T_n = n^{-1} \sum_{i=1}^n x_i$, then
$$v_F(T_n) = \frac{\sigma^2}{n} = \frac{\int x^2 \, \mathrm{d}\, F(x) - (\int x \, \mathrm{d}\, F(x))^2}{n},$$
which clearly is a function of F.

- If $\hat{F}_n$ is a consistent estimator of F, $v_{\hat{F}_n}(T_n)$ should be a consistent estimator of $v_F(T_n)$.
- Since $v_{\hat{F}_n}(T_n)$ may be difficult to compute, we approximate it with a simulation estimate denoted by $v_{boot}(T_n)$.
- Here is the bootstrap variance estimation procedure:
  1. Draw $x_1^*, \ldots, x_n^* \sim \hat{F}_n$;
  2. Compute $T_n^* = g(x_1^*, \ldots, x_n^*)$;
  3. Repeat steps 1 and 2, B times to get $T_{n,1}^*, \ldots, T_{n,B}^*$;
  4. Let
$$v_{boot} = B^{-1} \sum_{b=1}^{B} (T_{n,b}^* - B^{-1} \sum_{r=1}^{B} T_{n,r}^*)^2.$$
- By the LLN, $v_{boot} \to_{a.s.} v_{\hat{F}_n}(T_n)$ as $B \to \infty$.
- By the convergence of $\hat{F}_n \to F$, we expect $v_{\hat{F}_n}(T_n) \to v_F(T_n)$. (The proof mainly uses the nonparametric delta method.)

- One natural estimator $\hat{F}_n$ is the *empirical cumulative distribution function* (ECDF). It is the CDF that puts mass $1/n$ at each data point x_i,

$$\hat{F}_n(x) = n^{-1} \sum_{i=1}^{n} 1(x_i \leq x).$$

Theorem 3 (Glivenko-Cantelli Theorem) *Let $x_1, \ldots, x_n$ be random draws from F and let $\hat{F}_n$ be the ECDF. Then*

$$\sup_x \left| \hat{F}_n(x) - F(x) \right| \to_{a.s.} 0.$$

- Using the ECDF as $\hat{F}_n$, drawing n points at random from $\hat{F}_n$ is the same as drawing a sample of size n with replacement from the original data. Hence the first step can be replaced by "1. Draw $x_1^*, \ldots, x_n^*$ with replacement from $x_1, \ldots, x_n$."

Median estimation example

- Suppose we have iid draws $u_1, \ldots, u_n$ with n being odd from the uniform distribution $U[0, a]$. Letting $k = (n+1)/2$, it can be shown that

$$\mathrm{Var}(\widehat{median}(u_1, \ldots, u_n)) = \frac{a^2}{4 \cdot (2k+1)}.$$

- Here is an R program illustrating the bootstrap estimation of the variance.

```
set.seed(1011)
N <- 21
k <- (N + 1) / 2
u <- runif(n = N, min = 0, max = 10)
# theoretical sd
sd0 <- sqrt(10^2 / (4 * (2 * k + 1)))
# bootstrap
B <- 50
medianBootstrap <- c()
for (b in 1 : B) {
# draw u with replacement
ustar <- sample(u, size = N, replace = TRUE)
medianBootstrap[b] <- median(ustar)
}
sdbootstrap <- sd(medianBootstrap)
sdbootstrap
# 1.170141
sd0
# 1.042572
```

Inference

Single restriction

- To test single restriction like $H_0 : \theta_{0j} = a_j$. We can use

$$t(\hat{\theta}_j, a_j) = \frac{\hat{\theta}_j - a_j}{\mathrm{se}(\hat{\theta}_j)}.$$

Because $(\hat{\theta}_j - \theta_{0j})/\mathrm{se}(\hat{\theta}_j) \to_d \mathcal{N}(0,1)$, where the standard errors are obtained as the square roots of the diagonal elements of the asymptotic variance matrix, $t(\hat{\theta}_j, a_j) \to_d \mathcal{N}(0,1)$ under the null.

Multiple, nonlinear restrictions: Wald statistic

- We can test multiple, nonlinear restrictions using the asymptotic normality of $\hat{\theta}_n$ and delta method. Consider

$$H_0 : c(\theta_0) = 0,$$

where $c : \Theta \mapsto \mathbb{R}^q$. Assume c is continuously differentiable on $\mathrm{int}(\Theta)$. Let $C(\theta) = \nabla_\theta c(\theta)$ be the $q \times p = \dim(\theta)$ Jacobian matrix. Define $C \equiv C(\theta_0)$.

- By the mean value theorem argument,

$$\begin{aligned}\sqrt{n}(c(\hat{\theta}_n) - c(\theta_0)) &= \nabla_\theta c(\theta_0)\sqrt{n}(\hat{\theta}_n - \theta_0) + o_p(1) \\ &= C\sqrt{n}(\hat{\theta}_n - \theta_0) + o_p(1).\end{aligned}$$

- Under the null,

$$\sqrt{n}c(\hat{\theta}_n) \to_d \mathcal{N}(0, \Sigma \equiv CA^{-1}BA^{-1}C').$$

$$\sqrt{n}c(\hat{\theta}_n)'\Sigma^{-1}\sqrt{n}c(\hat{\theta}_n) \to_d \chi^2_q.$$

- The Wald statistic is obtained from replacing A, B and C with their consistent estimators,

$$W = nc(\hat{\theta}_n)' \left(\hat{C} \hat{A}^{-1} \hat{B} \hat{A}^{-1} \hat{C}' \right)^{-1} c(\hat{\theta}_n),$$

  where $\hat{C} = C(\hat{\theta}_n)$.
- Under the null, $W \to_d \chi^2_q$.
- The Wald relies on the estimation of the unrestricted model. It is easily available.

Score statistic or Lagrange multiplier (LM) statistic

- Unlike Wald, score statistic is based on restricted estimate. Let $\tilde{\theta}_n$ be the restricted estimate from

$$\min_{\theta \in \Theta} \sum_{i=1}^{n} q(z_i, \theta) \quad s.t. \quad c(\theta) = 0.$$

- Under the null, we have the mean value expansion for the score,

$$n^{-1/2} \sum_{i=1}^{n} s(z_i, \tilde{\theta}_n) = n^{-1/2} \sum_{i=1}^{n} s(z_i, \theta_0) + A\sqrt{n}(\tilde{\theta}_n - \theta_0) + o_p(1)$$

  Also,

$$\sqrt{n} c(\tilde{\theta}_n) = \sqrt{n} c(\theta_0) + C\sqrt{n}(\tilde{\theta}_n - \theta_0) + o_p(1).$$

- Note $c(\theta_0) = 0$ by the null and $c(\tilde{\theta}_n) = 0$ by the FOC of the Lagrange multiplier. We conclude that $C\sqrt{n}(\tilde{\theta}_n - \theta_0) = o_p(1)$.

- Hence, under the null,

$$CA^{-1}\left(n^{-1/2}\sum_{i=1}^{n} s(z_i,\tilde{\theta}_n)\right) = CA^{-1}\left(n^{-1/2}\sum_{i=1}^{n} s(z_i,\theta_0)\right) + o_p(1).$$

- Recall $n^{-1/2}\sum_{i=1}^{n} s(z_i,\theta_0) \to_d \mathcal{N}(0,B)$. Hence

$$CA^{-1}\left(n^{-1/2}\sum_{i=1}^{n} s(z_i,\tilde{\theta}_n)\right) \to_d \mathcal{N}(0, CA^{-1}BA^{-1}C').$$

- Then the quadratic form

$$LM = \left(\sum_{i=1}^{n}\tilde{s}_i\right)' \tilde{A}^{-1}\tilde{C}'\left(\tilde{C}\tilde{A}^{-1}\tilde{B}\tilde{A}^{-1}\tilde{C}'\right)^{-1}\tilde{C}'\tilde{A}^{-1}\left(\sum_{i=1}^{n}\tilde{s}_i\right)n^{-1} \to_d \chi^2_q.$$

All tilde terms are evaluated at $\tilde{\theta}_n$.

Quasi-likelihood ratio (QLR) statistic

- QLR test statistic is based on the difference of the objective function evaluated at the restricted and unrestricted estimates,
- Let $\tilde{\theta}_n$ and $\hat{\theta}_n$ be the restricted and unrestricted estimates. Using a second-order Taylor expansion, we have

$$\sum_{i=1}^{n} q(z_i, \tilde{\theta}_n) - \sum_{i=1}^{n} q(z_i, \hat{\theta}_n) = \left(\sum_{i=1}^{n} s_i(\hat{\theta}_n) \right) (\tilde{\theta}_n - \hat{\theta}_n) + (1/2)(\tilde{\theta}_n - \hat{\theta}_n)' \left(\sum_{i=1}^{n} H_i(\theta_*) \right) (\tilde{\theta}_n - \hat{\theta}_n),$$

  where θ_* lies in the line segment connecting $\tilde{\theta}_n$ and $\hat{\theta}_n$.
- By the FOC, $\sum_{i=1}^{n} s_i(\hat{\theta}_n) = 0$, and we have

$$2 \left(\sum_{i=1}^{n} q(z_i, \tilde{\theta}_n) - \sum_{i=1}^{n} q(z_i, \hat{\theta}_n) \right) = (\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n))' A (\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n)) + o_p(1) \tag{2}$$

- As for $\sqrt{n}(\hat{\theta}_n - \tilde{\theta}_n)$, we use the mean value expansion of $\sum_{i=1}^n s_i(\hat{\theta}_n)$ at $\tilde{\theta}_n$,

$$0 = n^{-1/2} \sum_{i=1}^n s_i(\hat{\theta}_n) = n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n) + A\sqrt{n}(\hat{\theta}_n - \tilde{\theta}_n) + o_p(1).$$

  Hence

$$\sqrt{n}(\hat{\theta}_n - \tilde{\theta}_n) = A^{-1} n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n) + o_p(1).$$

  Thus,

$$\text{eq. (2)} = \left(n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n) \right)' A^{-1} \left(n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n) \right) + o_p(1). \qquad (3)$$

- If $A = B$ (*generalized information matrix equality*), eq. (3), hence *QLR*, has the asymptotic distribution of χ^2_q. (See page 424 of Wooldridge for some matrix algebra arguments.)

Two-step estimation

Issues of two-step estimations

- A two-step estimator is one that depends on some preliminary, "first-step" estimator of parameters. The "first-step" can also be two or more step estimation.
- Examples include weighted OLS/NLS, efficient GMM estimation (the first step is about the weighting matrix), generated regressors.
- Question is whether the estimation of the first step affects the asymptotic variance of the second, and if so, what effect does the first step have.
- Ignoring the sampling error in the first step estimation might lead to incorrect confidence interval of the second step estimates.

Set up and consistency of two-step M estimation

- Let $\hat{\gamma}$ be a consistent estimator of a J dimensional parameter γ_*. A two-step M estimator $\hat{\theta}_n$ solves

$$\min_{\theta \in \Theta} n^{-1} \sum_{i=1}^{n} q(z_i, \theta; \hat{\gamma}).$$

- Under regularity conditions, including continuity of q in θ and γ, the key condition for consistency of $\hat{\theta}_n$ is that θ_0 uniquely solves

$$\min_{\theta \in \Theta} \mathrm{E}(q(z, \theta; \gamma_*)).$$

- This follows from using lemma 2.
- Asymptotic variance is more interesting.

Example: weighted NLS

- Suppose we specify $\mathrm{Var}(y|x) = h(x, \gamma) > 0$. We have a $\hat{\gamma} \to_p \gamma_*$ for some γ_*. For example, let $\tilde{u}_i = y_i - m(x_i, \tilde{\theta})$, where $\tilde{\theta}$ is the NLS estimate. Then $\hat{\gamma}$ solves

$$\min_{\gamma \in \Gamma} n^{-1} \sum_{i=1}^{n} (\tilde{u}_i^2 - h(x_i, \gamma))^2$$

- The weighted NLS (WNLS) estimator solves

$$\min_{\theta \in \Theta} n^{-1} \sum_{i=1}^{n} (y_i - m(x_i, \theta))^2 / h(x_i, \hat{\gamma}).$$

  If $h(x, \gamma)$ is a "good" approximation to $\mathrm{Var}(y|x)$, WNLS might be more efficient than NLS.

- Using lemma 2, we have

$$n^{-1} \sum_{i=1}^{n} (y_i - m(x_i, \theta))^2 / h(x_i, \hat{\gamma}) \to_p \mathrm{E}((y_i - m(x_i, \theta))^2 / h(x_i, \gamma_*)).$$

- If θ_0 uniquely minimizes $\mathrm{E}((y_i - m(x_i, \theta))^2 / h(x_i, \gamma_*))$, $\hat{\theta}_n$ is consistent for θ_0. We can show this.

- Because $\mathrm{E}(y_i - m(x_i, \theta_0) \mid x_i) = 0$, we conclude that for any x_i, θ_0 solves

$$\min_{\theta \in \Theta} \mathrm{E}\left((y_i - m(x_i, \theta))^2 \,\middle|\, x_i \right)$$

- Because $h(x, \gamma) > 0$ and is a function of x, we have

$$\begin{aligned} \mathrm{E}((y - m(x, \theta_0))^2 / h(x, \gamma) \mid x) &= \mathrm{E}((y - m(x, \theta_0))^2 \mid x) / h(x, \gamma) \\ &\leq \mathrm{E}((y - m(x, \theta))^2 \mid x) / h(x, \gamma). \end{aligned}$$

  Applying the law of iterated expectations, for any θ and any γ,

$$\mathrm{E}((y - m(x, \theta_0))^2 / h(x, \gamma)) \leq \mathrm{E}((y - m(x, \theta))^2 / h(x, \gamma)).$$

- We then just need to assume uniqueness of θ_0.
- WNLS consistently estimate θ_0 for any positive weighting function (including misspecified ones).

Asymptotic variance of two-step M estimation

- Suppose
$$\sqrt{n}(\hat{\gamma} - \gamma_*) = n^{-1/2} \sum_{i=1}^{n} r_i(\gamma_*) + o_p(1),$$
where $\mathrm{E}(r_i(\gamma_*)) = 0$.
- $r_i(\gamma_*)$ is called "influence function". It describes completely the first order asymptotic properties of $\hat{\gamma}$.
- Usually, $r_i(\gamma_*)$ takes the form $Mh(\gamma_*)$ for some matrix M and function h. For example, if the $\hat{\gamma}$ is an M estimator, we know that
$$\sqrt{n}(\hat{\gamma} - \gamma_*) = A^{-1} n^{-1/2} \sum_{i=1}^{n} s_i(\gamma_*) + o_p(1), \quad A = \mathrm{E}(H(\gamma_*)),$$
where s_i is certain score function, and H is certain Hessian matrix. Then $r_i(\gamma_*) = A^{-1} s_i(\gamma_*)$.

- By the arguments of M estimation, we have the expansion,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = A^{-1}\left(-n^{-1/2}\sum_{i=1}^{n} s(z_i, \theta_0; \hat{\gamma})\right) + o_p(1),$$

  where $s(z_i, \theta; \gamma)$ is the score $\nabla_\theta q(z_i, \theta; \gamma)$ with respect to θ, and A is the expected Hessian matrix

$$A = \mathrm{E}(\nabla_\theta s(z, \theta_0; \gamma_*)).$$

- The score $s(z_i, \theta_0; \hat{\gamma})$ still contains sampling error.

- Letting $s_i(\theta_0, \hat{\gamma}) = s(z_i, \theta_0; \hat{\gamma})$, we have

$$
\begin{aligned}
n^{-1/2} \sum_{i=1}^{n} s_i(\theta_0, \hat{\gamma}) &= n^{-1/2} \sum_{i=1}^{n} s_i(\theta_0, \gamma_*) + F\sqrt{n}(\hat{\gamma} - \gamma_*) + o_p(1) \\
&= n^{-1/2} \sum_{i=1}^{n} s_i(\theta_0, \gamma_*) + Fn^{-1/2} \sum_{i=1}^{n} r_i(\gamma_*) + o_p(1) \\
&= n^{-1/2} \sum_{i=1}^{n} \Big(s_i(\theta_0, \gamma_*) + Fr_i(\gamma_*) \Big) + o_p(1)
\end{aligned}
$$

where

$$
F = \mathrm{E}(\nabla_\gamma s_i(\theta_0, \gamma_*)).
$$

- Combining the above results, we have

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = A^{-1}\left(-n^{-1/2}\sum_{i=1}^{n}\big(s_i(\theta_0,\gamma_*) + Fr_i(\gamma_*)\big)\right) + o_p(1)$$
$$= A^{-1}\left(-n^{-1/2}\sum_{i=1}^{n} g_i(\theta_0,\gamma_*)\right) + o_p(1),$$

where

$$g_i(\theta_0,\gamma_*) = s_i(\theta_0,\gamma_*) + Fr_i(\gamma_*).$$

- Note $\mathrm{E}(g_i(\theta_0,\gamma_*)) = 0$. We can apply the CLT to obtain

$$-n^{-1/2}\sum_{i=1}^{n} g_i(\theta_0,\gamma_*) \to \mathcal{N}(0,D), \quad D = \mathrm{E}(g_i(\theta_0,\gamma_*)g_i(\theta_0,\gamma_*)').$$

We conclude that

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \to_d \mathcal{N}(0, A^{-1}DA^{-1}).$$

Discussion about the asymptotic variance

- The asymptotic variance of the two-step M estimator $\hat{\theta}_n$ is

$$n^{-1}A^{-1}DA^{-1},$$

where

$$\begin{aligned} D &= \mathrm{E}\big(g_i(\theta_0,\gamma_*)g_i(\theta_0,\gamma_*)'\big) \\ &= \mathrm{E}\big(s_i(\theta_0,\gamma_*)s_i(\theta_0,\gamma_*)'\big) + \mathrm{E}\big(s_i(\theta_0,\gamma_*)r_i(\gamma_*)'\big)F' \\ &\quad + F\,\mathrm{E}\big(r_i(\gamma_*)s_i(\theta_0,\gamma_*)'\big) + F\,\mathrm{E}\big(r_i(\gamma_*)r_i(\gamma_*)'\big)F'. \end{aligned}$$

- Let $B = \mathrm{E}(s_i(\theta_0,\gamma_*)s_i(\theta_0,\gamma_*)')$. If we ignore the sampling error in the first step estimation, the variance will be

$$n^{-1}A^{-1}BA^{-1}.$$

- In some cases, $F = \mathrm{E}(\nabla_\gamma s(z_i, \theta_0, \gamma_*))$ is zero. So it is legitimate to ignore the estimation error in $\hat{\gamma}$.
- In some cases, $s_i(\theta_0, \gamma_*)$ and $r_i(\gamma_*)$ are uncorrelated, that is $\mathrm{E}(s_i r_i') = 0$. Then the variance ignoring the first step estimation error will be too small.
- For asymptotic variance of two-step semiparametric estimators, see Newey (1994, The Asymptotic variance of semiparametric estimators, Econometrica).

Estimating the asymptotic variance of the two-step estimator

- Let
$$\hat{F} = n^{-1} \sum_{i=1}^{n} \nabla_\gamma s_i(\hat{\theta}_n, \hat{\gamma}) \to_p F, \quad \hat{g}_i = s_i(\hat{\theta}_n, \hat{\gamma}) + \hat{F} r_i(\hat{\gamma}).$$
- Because $D = \mathrm{E}(g_i(\theta_0, \gamma_*) g_i(\theta_0, \gamma_*)')$, D can be estimated by
$$n^{-1} \sum_{i=1}^{n} g_i(\theta_0, \gamma_*) g_i(\theta_0, \gamma_*)'.$$
- Then
$$\hat{D} = n^{-1} \sum_{i=1}^{n} \hat{g}_i \hat{g}_i' \to_p D.$$
- $\hat{A} = n^{-1} \sum_{i=1}^{n} H_i(\hat{\theta}_n, \hat{\gamma}) \to_p A$, where $H_i(\hat{\theta}_n, \hat{\gamma}) = \nabla_\gamma s(z_i, \hat{\theta}_n; \hat{\gamma})$.
- A valid variance estimator is $n^{-1} \hat{A}^{-1} \hat{D} \hat{A}^{-1}$.

WNLS example

- In the WNLS case, we have the Hessian

$$\nabla_\gamma s_i(\theta,\gamma) = \nabla_\theta m(x_i,\theta)' \nabla_\theta h(x,\gamma) u_i(\theta)/(h(x,\gamma)^2)$$

  for any γ.
- Note $\mathrm{E}(u_i(\theta_0) \mid x_i) = 0$. We have $F = \mathrm{E}(\nabla_\gamma s_i(\theta_0,\gamma_*)) = 0$.
- There is no need to adjust the variance of $\hat{\theta}_n$ for the two-step WNLS.
- Thus, we just use $n^{-1}\hat{A}^{-1}\hat{B}^{-1}\hat{A}^{-1}$ as the asymptotic variance estimator,

$$\left(\sum_{i=1}^n \nabla_\theta \hat{m}_i' \nabla_\theta \hat{m}_i/\hat{h}_i\right)^{-1} \left(\sum_{i=1}^n \hat{u}_i^2 \nabla_\theta \hat{m}_i' \nabla_\theta \hat{m}_i/\hat{h}_i^2\right) \left(\sum_{i=1}^n \nabla_\theta \hat{m}_i' \nabla_\theta \hat{m}_i/\hat{h}_i\right)^{-1}$$

  Here $\hat{m}_i = m(x_i,\hat{\theta}_n)$ and $\hat{h}_i = h(x_i,\hat{\gamma})$.